Structure and Physical state

The vegetable oil is a liquid at room temperature. However, its hydrogenated form (Vanaspathi) is a solid. Do you know why?

The vegetable oil contains mostly unsaturated fatty acids while the Vanaspathi, the hydrogenated vegetable oil consists mainly saturated fatty acids. Saturated fatty acids are fairly linear and pack together closely through hydrophobic interactions. Close packing of saturated fats promotes stability and causes saturated fats to form solids at room temperature. As unsaturated fatty acids (especially with cis-configuration) form kinked structures as shown in the picture, close packing of unsaturated fats is prevented. Hence, these are primarily liquids at room temperature.

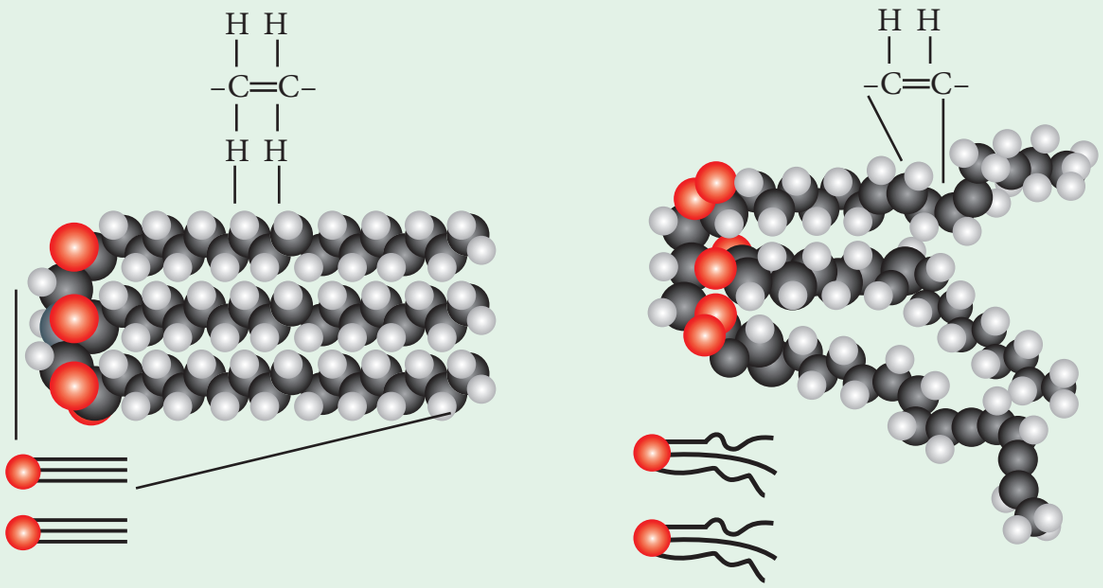


Table 6.1 & 6.2 Show the symbol, structure and common name of saturated, unsaturated fatty acid, essential fatty acid and the unsaturated trans fatty acid, elaidic acid.

Table 6.1 Saturated fatty acids

Symbol	Structure	Systemic name	Common name
12:0	$CH_3(CH_2)10COOH$	*n*-Dodecanoic acid	Lauric acid
16:0	$CH_3(CH_2)14COOH$	*n*-Hexadecanoic acid	Palmitic acid
18:0	$CH_3(CH_2)16\ COOH$	*n*-Octadecanoic acid	Stearic acid

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